

Engineering encyclopedia

A request-level technical benchmark of 11 DigitalOcean-hosted endpoints across latency, throughput, context, output, tools, structured output, vision, quality, overload, and recovery. A historical partner-model mistake is isolated in an incident appendix and excluded from production comparisons.

11	40,049	176/176	\$329.972
DO-HOSTED ENDPOINTS	NORMALIZED REQUESTS	CONCLUSIVE COVERAGE CELLS	CONSERVATIVE EXPOSURE

How to read this report: every number is tied to one endpoint and one workload. Intervals describe the measured sample—not a provider guarantee. An arrow or ‘lower bound’ means the endpoint did not fail before the test stopped. ‘Inconclusive’ is never silently converted to supported or unsupported.

1. What the evidence says

The completed runs show that these endpoints can serve substantial workloads, but they are not interchangeable and should not be operated from one global RPM number. Throughput depends strongly on whether the bottleneck is request rate, prompt prefill, decoding, or a heterogeneous mix. The measured cells are engineering anchors—not current production authorization or contractual quotas.

Inference balance gate · restored, verify before load

The interrupted closure wave encountered HTTP 403 while the Serverless Inference prepaid balance was depleted. After the owner replenished the balance, the same inference credential again returned HTTP 200 from /v1/models. The separate account-control endpoint still returns 403 and is not used as an inference-readiness gate. Run two cheap serial DO-hosted marker controls before each new load wave.

40,049	1,343	176	0
REQUEST OBSERVATIONS	LOAD EPOCHS	RESOLVED CELLS	OPERATIONAL FAILURES

Production decision rules

Observed condition	Engineering action
HTTP 429	Honor reset/retry hints, multiplicatively decrease offered load, then additively recover.
HTTP 5xx or timeout	Retry with bounded exponential backoff and jitter; count the failed attempt in reliability and cost.
HTTP 400 validation error	Do not retry unchanged. Correct the exact parameter or payload state.
HTTP 401/403 on controls	Stop the lane. Treat it as credential/account access failure, not a model capability result.
AIMD lower bound only	Treat the number as a tested floor, not a quota or sustainable guarantee.
Two-minute cell measured but composite failed	Use the transport/latency measurements, but do not promote the cell as production-verified without resolving the failed criterion.
Capability inconclusive	Feature-gate it. Do not infer support from model-family marketing or a generic 2xx response.

Attribution boundary

These measurements characterize DigitalOcean's serving route plus the named model. No identical external-provider control was used, so a quality or latency difference cannot be assigned uniquely to the base model, quantization, serving stack, or routing layer.

Production triage

No heterogeneous mixed-load cell passed the full soak contract

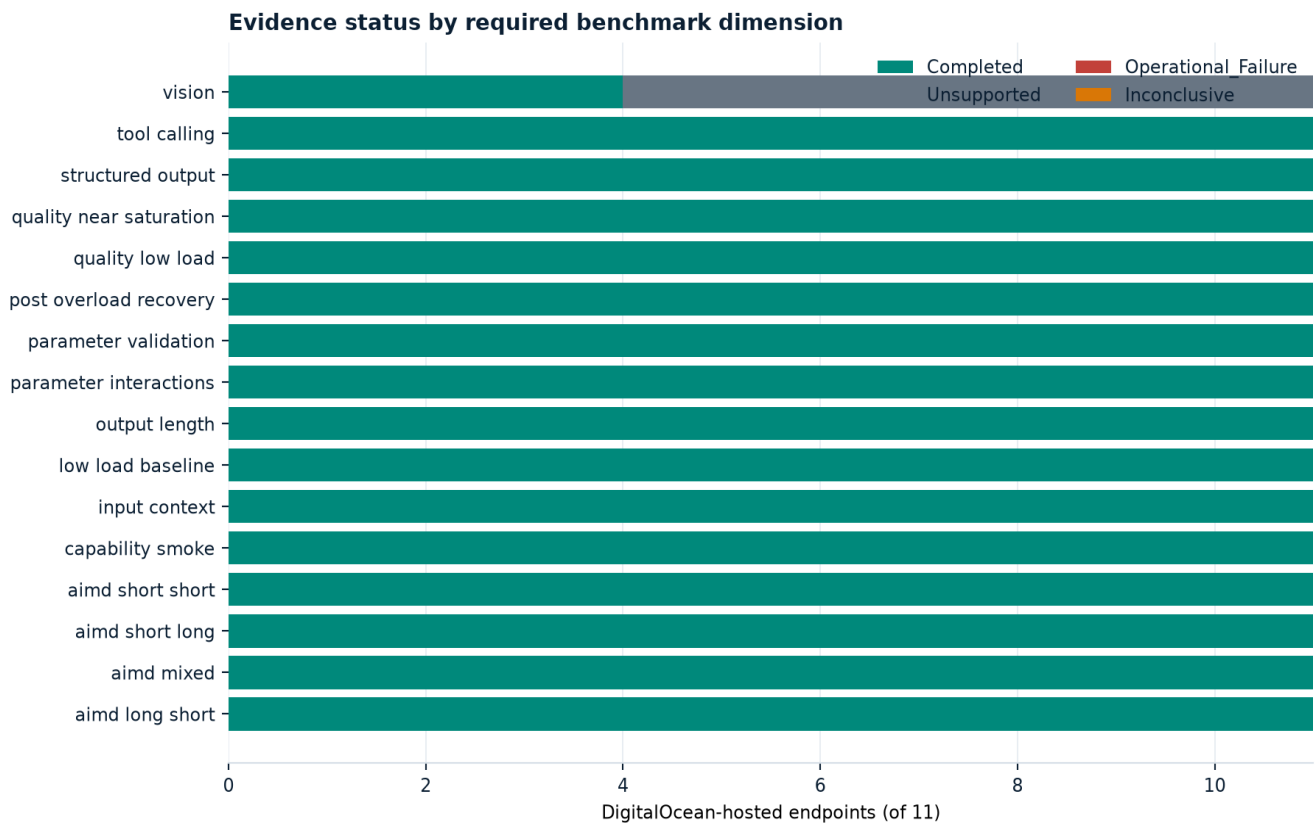
The campaign measured all hosted endpoint × workload cells, but a completed measurement is not automatically a production pass. Mixed traffic should remain behind workload-matched canaries until its own two-minute acceptance, quality, and recovery criteria pass.

Decision lane	Endpoints	Evidence-backed action
Guarded short/short pilot	DeepSeek V4 Flash 0731, Gemma 4 31B IT, Qwen 3.8 Max	These exact 1 RPS cells passed the strict two-minute composite. Start below the measured point and revalidate; do not generalize to other shapes.
Hold tool-dependent traffic	Nemotron 3 Super, Qwen 3.5 397B	Matched tool probes were transport-degraded. Adjacent controls succeeded, so feature-gate tools rather than retrying blindly.
Functionally verified vision	Gemma 4 31B IT, Qwen 3.8 Max	Only these routes both accepted the image request and passed the deterministic vision check in this campaign.
≈1M-token retrieval verified	Qwen 3.8 Max	Only this route recovered the embedded retrieval marker at roughly one million realized prompt tokens.

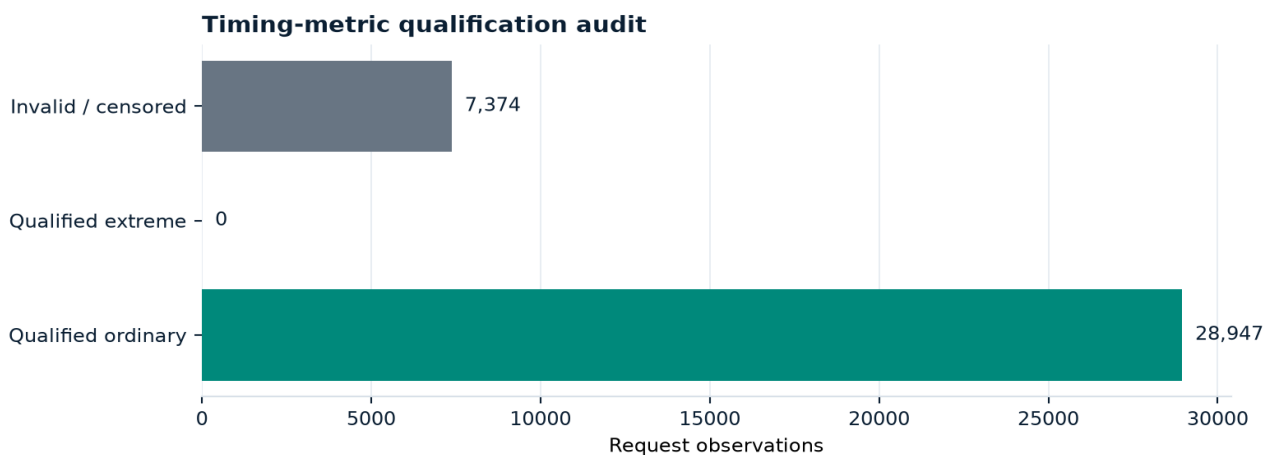
How to interpret the red flags

A transport-degraded feature is an operational route failure for that exact request contract, not proof that the underlying model family lacks the feature everywhere. Likewise, HTTP acceptance at long context is weaker than correct retrieval. The endpoint profiles later in this report preserve the exact rate, workload, sample size, interval, and censoring state behind each decision.

2. Coverage and measurement quality



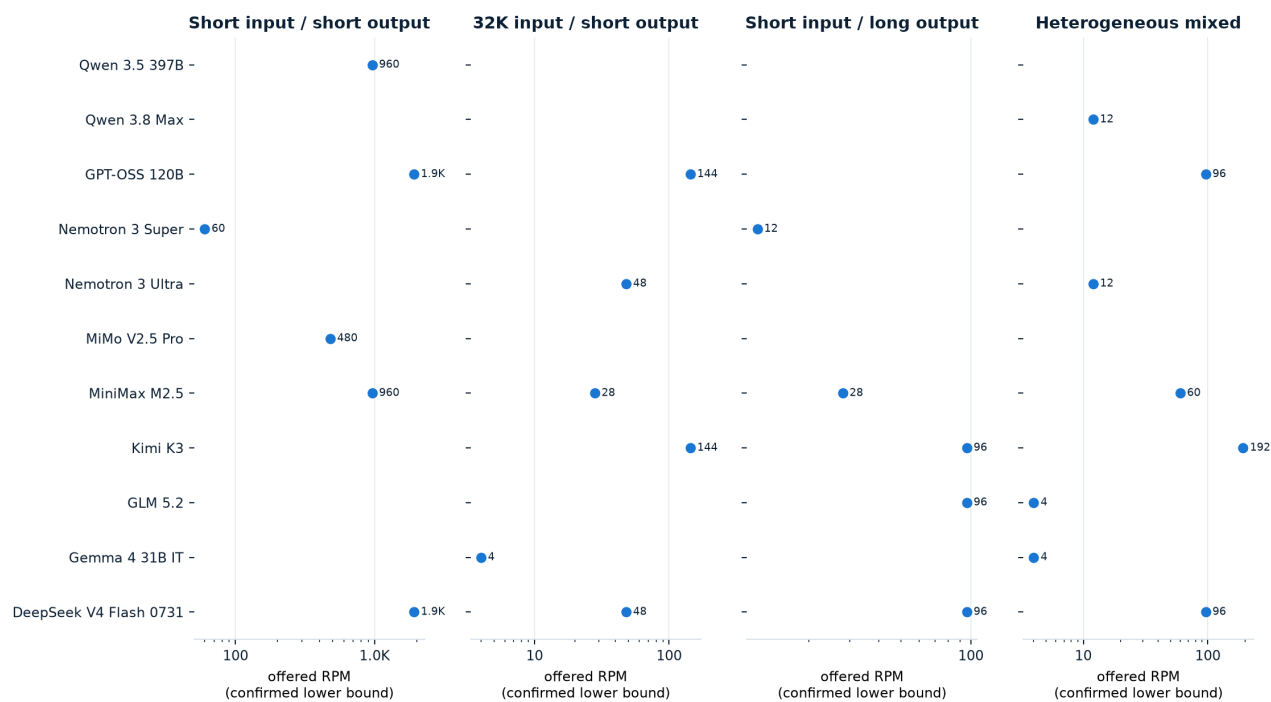
Completed means the planned cell produced interpretable evidence. Unsupported requires documentation or an exact evidence-backed rejection. Inconclusive includes transport failures, missing prerequisites, or insufficient statistical confirmation; it is not a euphemism for failure.



The timing audit removes buffered TTFT, sub-100 ms unstable post-TTFT denominators, multi-choice aggregate usage, and other invalid observations before any tokens/second summary. Qualified extremes remain visible and flagged; nothing is winsorized or silently clipped.

3. AIMD capacity map

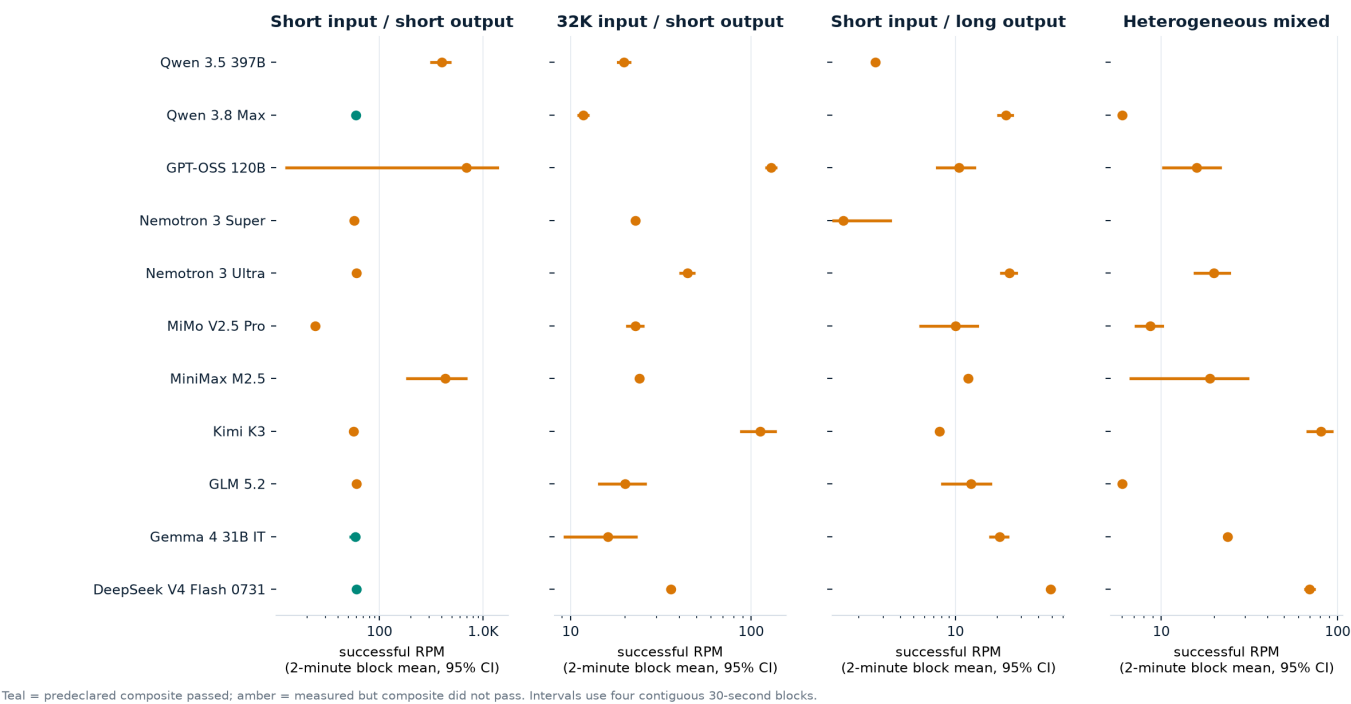
AIMD healthy offered-rate lower bounds — matched workload cells



Each dot is the highest endpoint/workload offered rate with the required separated healthy confirmations. It is a lower bound when no overload knee was reached.

4. Two-minute goodput

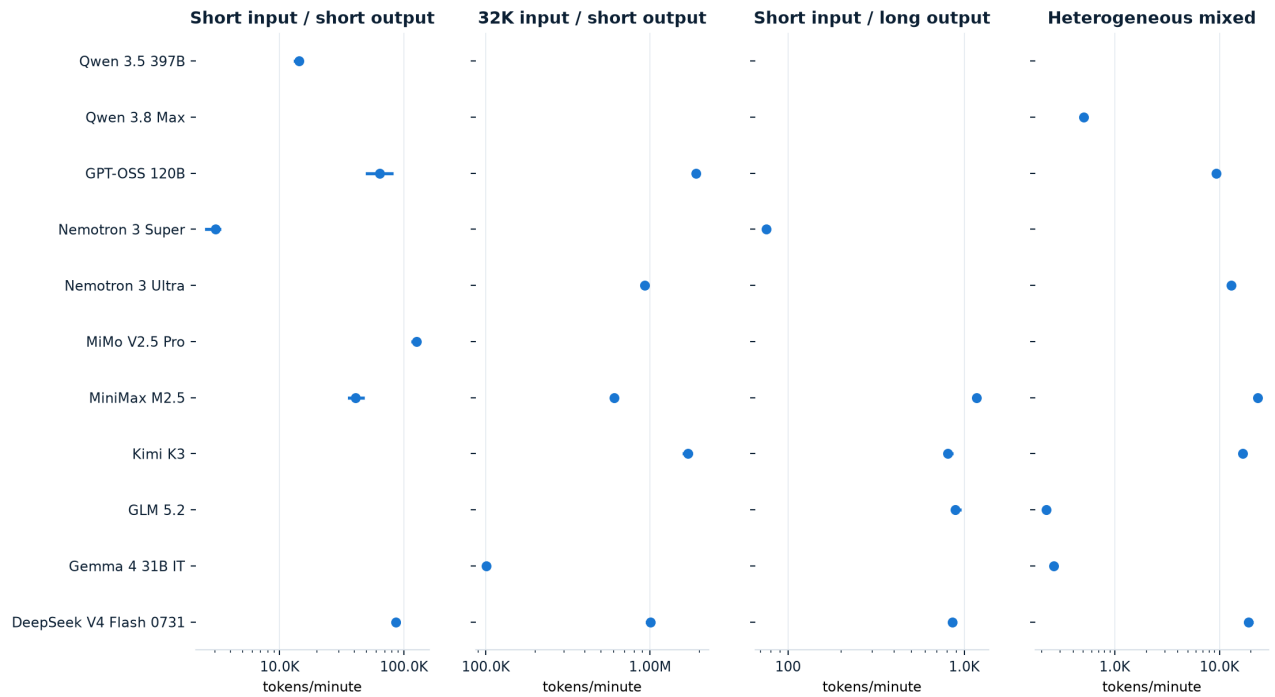
Two-minute achieved goodput at the tested candidate rate



Dots and intervals summarize four predeclared 30-second blocks. The interval is exploratory because adjacent blocks may be serially correlated.

5. Input-token goodput

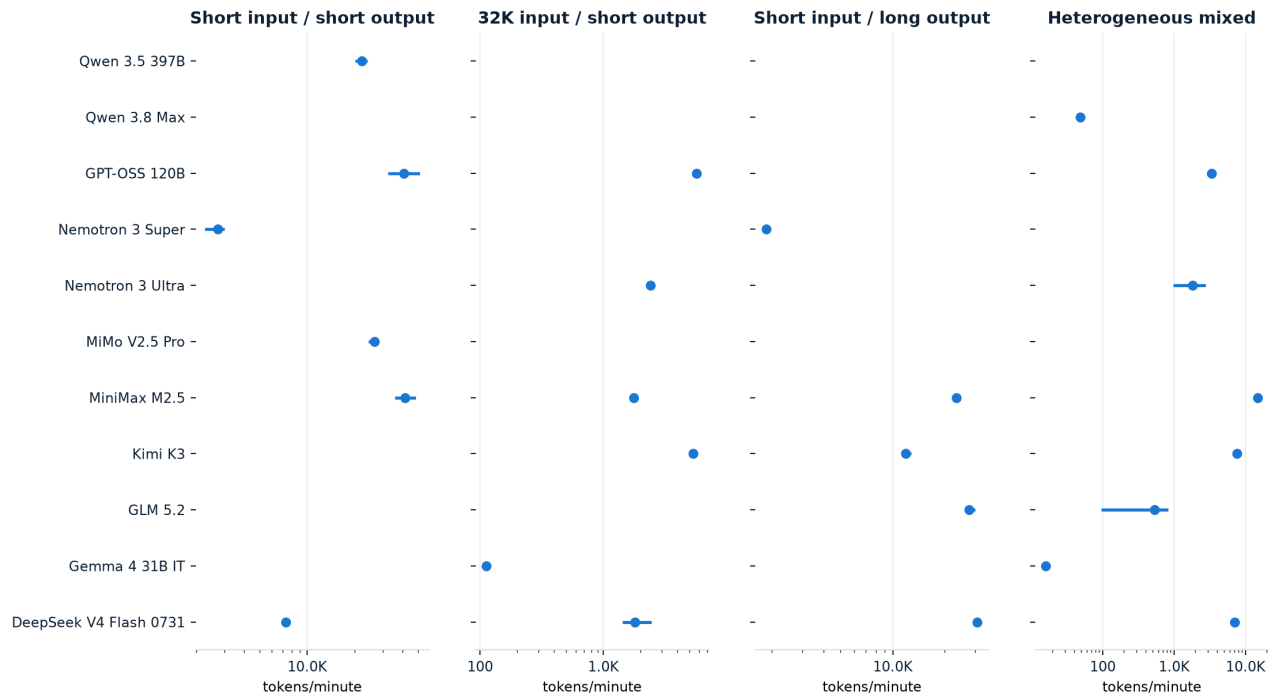
Effective input throughput at the confirmed AIMD cell



Successful prompt tokens divided by complete elapsed wall time. This is end-to-end input goodput—not direct server-side prefill speed.

6. Output-token goodput

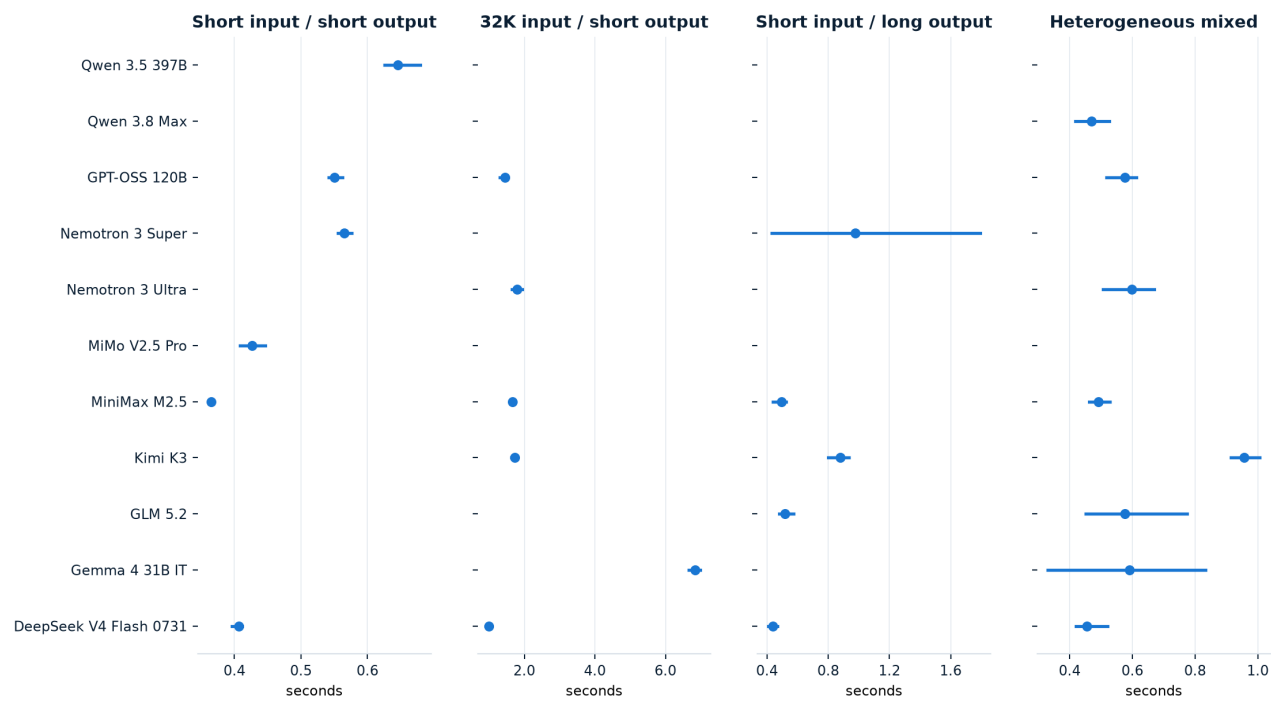
Effective output throughput at the confirmed AIMD cell



Successfully generated completion tokens divided by complete elapsed wall time, including queueing and drain.

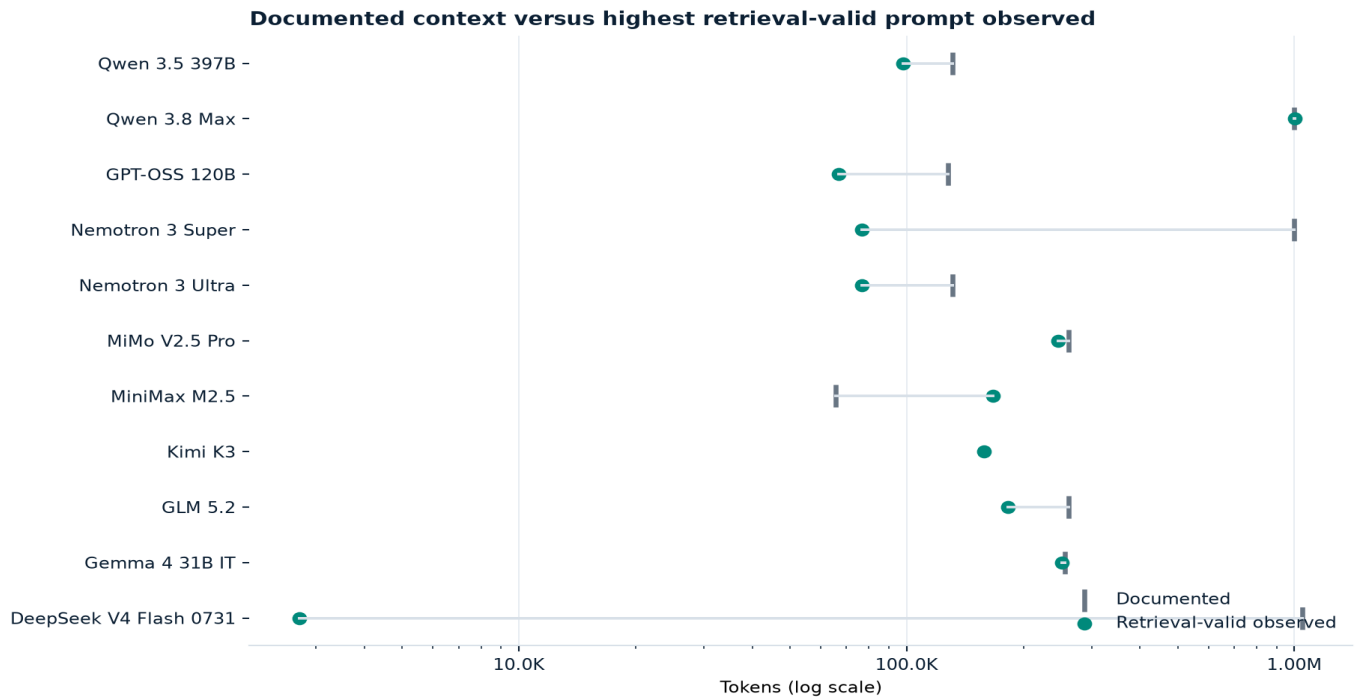
7. Time to first streamed token

Median time to first streamed token at the confirmed AIMD cell



TTFT is reported only when streaming made the first content event observable. Buffered responses are censored rather than mislabeled.

8. Context envelope



Retrieval-valid means a marker embedded in synthetic long context was recovered correctly. Mere HTTP acceptance is not counted as functional long-context success.

Endpoint 01 · DeepSeek V4 Flash 0731

deepseek-v4-flash-0731

9.8K	136	39	16/16
REQUESTS	LOAD EPOCHS	ERRORS	RESOLVED DIMENSIONS

Inventory

Provider	API	Documented context	Documented output	Input \$/M	Output \$/M
DeepSeek	chat_completions / v1	1.05M	1.05M	\$0.080	\$0.252

Documented modalities: text. Documented tools: None. Structured output: None. Prompt caching: True.

Measured operating anchors

Workload	AIMD healthy offered	Epochs	2-min achieved RPM (95% CI)	2-min status
Short input / short output	≥ 1.9K RPM	3	60.0 [60.0, 60.0]	composite passed
32K input / short output	≥ 48 RPM	3	36.0 [36.0, 36.0]	measured; composite failed
Short input / long output	≥ 96 RPM	3	48.8 [46.2, 51.5]	measured; composite failed
Heterogeneous mixed	≥ 96 RPM	3	69.4 [65.5, 73.2]	measured; composite failed

Interpretation: the AIMD column is the highest confirmed healthy offered-rate lower bound in the matched workload. The two-minute column is achieved goodput, including drain. Neither is a contractual quota or a diurnal guarantee.

DeepSeek V4 Flash 0731 · capability and limits

Capability and validation evidence

Capability / parameter	Transport	Functional	Valid n	Malformed validation
vision	observed transport degraded	not scored	1	not tested
tools	observed supported	passed	2	not tested
parallel tool calls	observed supported	passed	2	not tested
temperature	observed supported	passed	7	not tested
top p	observed supported	passed	7	not tested
stop	observed supported	passed	1	not tested
seed	observed supported	passed	2	not tested
caching option	documented unavailable	not scored	0	not tested
capability smoke	observed supported	passed	1	not tested
max completion tokens	observed supported	passed	1	not tested
max tokens	observed supported	degraded	3	correct rejection observed
pairwise core	observed supported	degraded	33	not tested
pairwise core	observed supported	degraded	7	not tested
parameter interaction extended	observed supported	failed	7	not tested
parameter interaction penalties	observed supported	passed	7	not tested
parameter interaction temperature top p output	observed supported	passed	6	not tested
top logprobs	observed supported	passed	7	not tested

Context and output envelope

Envelope	Documented	Observed functional / realized	Censoring
prompt context window	1.05M	2.7K	interval censored
combined prompt + requested output	1.05M	1.05M	interval censored
output limit	1.05M	10.0K	—
vision	—	—	—
tool calling	—	—	—
structured output	—	—	—

Deployment guidance

Current gate: /v1/models access recovered after the prepaid balance was replenished; pass two fresh serial streamed controls before deployment or another load wave. Then feature-gate from observed evidence, not the family name. Start below the relevant measured anchor, keep a separate concurrency ceiling, and use open-loop AIMD so rising latency cannot hide offered load. Back off on 429; retry bounded 5xx/timeouts; never retry an unchanged validation error. Re-run the exact profile after a model version, region, quota, or serving-stack change.

Endpoint 02 · Gemma 4 31B IT

gemma-4-31B-it

1.1K	112	108	16/16
REQUESTS	LOAD EPOCHS	ERRORS	RESOLVED DIMENSIONS

Inventory

Provider	API	Documented context	Documented output	Input \$/M	Output \$/M
Google	chat_completions / v1	256.0K	8.2K	\$0.180	\$0.500

Documented modalities: text. Documented tools: None. Structured output: None. Prompt caching: True.

Measured operating anchors

Workload	AIMD healthy offered	Epochs	2-min achieved RPM (95% CI)	2-min status
Short input / short output	—	—	58.2 [52.9, 63.4]	composite passed
32K input / short output	≥ 4 RPM	3	16.1 [9.3, 23.0]	measured; composite failed
Short input / long output	—	—	20.8 [17.9, 23.8]	measured; composite failed
Heterogeneous mixed	≥ 4 RPM	3	23.7 [23.1, 24.3]	measured; composite failed

Interpretation: the AIMD column is the highest confirmed healthy offered-rate lower bound in the matched workload. The two-minute column is achieved goodput, including drain. Neither is a contractual quota or a diurnal guarantee.

Gemma 4 31B IT · capability and limits

Capability and validation evidence

Capability / parameter	Transport	Functional	Valid n	Malformed validation
vision	observed supported	passed	1	not tested
tools	observed supported	passed	2	not tested
parallel tool calls	observed supported	passed	2	not tested
temperature	observed supported	passed	7	not tested
top p	observed supported	passed	7	not tested
stop	observed supported	passed	1	not tested
seed	observed supported	passed	2	not tested
caching option	documented unavailable	not scored	0	not tested
capability smoke	observed supported	passed	1	not tested
max completion tokens	observed supported	passed	1	not tested
max tokens	observed supported	degraded	3	invalid input accepted
pairwise core	observed supported	degraded	33	not tested
pairwise core	observed supported	degraded	7	not tested
parameter interaction extended	observed supported	failed	7	not tested
parameter interaction penalties	observed supported	passed	7	not tested
parameter interaction temperature top p output	observed supported	passed	6	not tested
top logprobs	observed supported	passed	7	not tested

Context and output envelope

Envelope	Documented	Observed functional / realized	Censoring
prompt context window	256.0K	251.9K	right censored
combined prompt + requested output	256.0K	126.0K	right censored
output limit	8.2K	28.8K	—
vision	—	—	—
tool calling	—	—	—
structured output	—	—	—

Deployment guidance

Current gate: /v1/models access recovered after the prepaid balance was replenished; pass two fresh serial streamed controls before deployment or another load wave. Then feature-gate from observed evidence, not the family name. Start below the relevant measured anchor, keep a separate concurrency ceiling, and use open-loop AIMD so rising latency cannot hide offered load. Back off on 429; retry bounded 5xx/timeouts; never retry an unchanged validation error. Re-run the exact profile after a model version, region, quota, or serving-stack change.

Endpoint 03 · GLM 5.2

glm-5.2

1.3K REQUESTS	114 LOAD EPOCHS	209 ERRORS	16/16 RESOLVED DIMENSIONS
-------------------------------------	---------------------------------------	----------------------------------	---

Inventory

Provider	API	Documented context	Documented output	Input \$/M	Output \$/M
Z.ai	chat_completions / v1	262.1K	262.1K	\$0.700	\$2.200

Documented modalities: text. Documented tools: True. Structured output: True. Prompt caching: True.

Measured operating anchors

Workload	AIMD healthy offered	Epochs	2-min achieved RPM (95% CI)	2-min status
Short input / short output	—	—	59.7 [59.0, 60.3]	measured; composite failed
32K input / short output	—	—	20.1 [14.4, 25.8]	measured; composite failed
Short input / long output	≥ 96 RPM	3	13.0 [8.1, 17.9]	measured; composite failed
Heterogeneous mixed	≥ 4 RPM	3	6.0 [5.9, 6.0]	measured; composite failed

Interpretation: the AIMD column is the highest confirmed healthy offered-rate lower bound in the matched workload. The two-minute column is achieved goodput, including drain. Neither is a contractual quota or a diurnal guarantee.

GLM 5.2 · capability and limits

Capability and validation evidence

Capability / parameter	Transport	Functional	Valid n	Malformed validation
vision	observed transport degraded	not scored	1	not tested
tools	observed supported	passed	2	not tested
parallel tool calls	observed supported	passed	2	not tested
temperature	observed supported	degraded	7	not tested
top p	observed supported	degraded	7	not tested
stop	observed supported	failed	1	not tested
seed	observed supported	failed	2	not tested
caching option	documented unavailable	not scored	0	not tested
capability smoke	observed supported	passed	1	not tested
max completion tokens	observed supported	failed	1	not tested
max tokens	observed supported	degraded	3	invalid input accepted
pairwise core	observed transport degraded	not scored	33	not tested
pairwise core	observed transport degraded	not scored	7	not tested
parameter interaction extended	observed transport degraded	not scored	7	not tested
parameter interaction penalties	observed supported	failed	7	not tested
parameter interaction temperature top p output	observed supported	degraded	6	not tested
response format	observed supported	failed	2	not tested
tool argument bytes	observed supported	failed	4	not tested

Context and output envelope

Envelope	Documented	Observed functional / realized	Censoring
prompt context window	262.1K	182.8K	right censored
combined prompt + requested output	262.1K	261.2K	right censored
output limit	262.1K	4.1K	—
vision	—	—	—
tool calling	—	—	—
structured output	—	—	—

Deployment guidance

Current gate: /v1/models access recovered after the prepaid balance was replenished; pass two fresh serial streamed controls before deployment or another load wave. Then feature-gate from observed evidence, not the family name. Start below the relevant measured anchor, keep a separate concurrency ceiling, and use open-loop AIMD so rising latency cannot hide offered load. Back off on 429; retry bounded 5xx/timeouts; never retry an unchanged validation error. Re-run the exact profile after a model version, region, quota, or serving-stack change.

Endpoint 04 · Kimi K3

kimi-k3

2.0K REQUESTS	117 LOAD EPOCHS	123 ERRORS	16/16 RESOLVED DIMENSIONS
-------------------------------------	---------------------------------------	----------------------------------	---

Inventory

Provider	API	Documented context	Documented output	Input \$/M	Output \$/M
Moonshot AI	chat_completions / v1	—	—	\$2.850	\$14.250

Documented modalities: text, image. Documented tools: None. Structured output: None. Prompt caching: True.

Measured operating anchors

Workload	AIMD healthy offered	Epochs	2-min achieved RPM (95% CI)	2-min status
Short input / short output	—	—	56.0 [53.1, 58.9]	measured; composite failed
32K input / short output	≥ 144 RPM	3	112 [88.2, 136]	measured; composite failed
Short input / long output	≥ 96 RPM	3	7.7 [7.6, 7.8]	measured; composite failed
Heterogeneous mixed	≥ 192 RPM	3	80.2 [67.5, 92.9]	measured; composite failed

Interpretation: the AIMD column is the highest confirmed healthy offered-rate lower bound in the matched workload. The two-minute column is achieved goodput, including drain. Neither is a contractual quota or a diurnal guarantee.

Kimi K3 · capability and limits

Capability and validation evidence

Capability / parameter	Transport	Functional	Valid n	Malformed validation
vision	observed supported	failed	18	correct rejection observed
tools	observed supported	failed	2	not tested
parallel tool calls	observed supported	degraded	2	not tested
temperature	observed supported	failed	7	not tested
top p	observed transport degraded	not scored	7	not tested
stop	observed supported	failed	1	not tested
seed	observed supported	degraded	2	not tested
caching option	documented unavailable	not scored	0	not tested
capability smoke	observed supported	passed	1	not tested
max completion tokens	observed supported	failed	1	not tested
max tokens	observed supported	degraded	3	invalid input accepted
pairwise core	observed transport degraded	not scored	33	not tested
pairwise core	observed transport degraded	not scored	7	not tested
parameter interaction extended	observed supported	failed	7	not tested
parameter interaction penalties	observed transport degraded	not scored	7	not tested
parameter interaction temperature top p output	observed transport degraded	not scored	6	not tested
top logprobs	observed supported	degraded	7	not tested

Context and output envelope

Envelope	Documented	Observed functional / realized	Censoring
prompt context window	—	158.4K	right censored
combined prompt + requested output	—	49.6K	right censored
output limit	—	4.1K	—
vision	—	—	—
tool calling	—	—	—
structured output	—	—	—

Deployment guidance

Current gate: /v1/models access recovered after the prepaid balance was replenished; pass two fresh serial streamed controls before deployment or another load wave. Then feature-gate from observed evidence, not the family name. Start below the relevant measured anchor, keep a separate concurrency ceiling, and use open-loop AIMD so rising latency cannot hide offered load. Back off on 429; retry bounded 5xx/timeouts; never retry an unchanged validation error. Re-run the exact profile after a model version, region, quota, or serving-stack change.

Endpoint 05 · MiniMax M2.5

minimax-m2.5

6.7K	155	676	16/16
REQUESTS	LOAD EPOCHS	ERRORS	RESOLVED DIMENSIONS

Inventory

Provider	API	Documented context	Documented output	Input \$/M	Output \$/M
MiniMax	chat_completions / v1	65.5K	65.5K	\$0.300	\$1.200

Documented modalities: text. Documented tools: None. Structured output: None. Prompt caching: True.

Measured operating anchors

Workload	AIMD healthy offered	Epochs	2-min achieved RPM (95% CI)	2-min status
Short input / short output	≥ 960 RPM	3	431 [187, 676]	measured; composite failed
32K input / short output	≥ 28 RPM	3	24.0 [24.0, 24.0]	measured; composite failed
Short input / long output	≥ 28 RPM	3	12.4 [12.2, 12.6]	measured; composite failed
Heterogeneous mixed	≥ 60 RPM	3	18.8 [6.7, 30.9]	measured; composite failed

Interpretation: the AIMD column is the highest confirmed healthy offered-rate lower bound in the matched workload. The two-minute column is achieved goodput, including drain. Neither is a contractual quota or a diurnal guarantee.

MiniMax M2.5 · capability and limits

Capability and validation evidence

Capability / parameter	Transport	Functional	Valid n	Malformed validation
vision	observed transport degraded	not scored	1	not tested
tools	observed supported	degraded	2	not tested
parallel tool calls	observed supported	degraded	2	not tested
temperature	observed supported	failed	7	not tested
top p	observed supported	failed	7	not tested
stop	observed supported	failed	1	not tested
seed	observed supported	failed	2	not tested
caching option	documented unavailable	not scored	0	not tested
capability smoke	observed supported	failed	1	not tested
max completion tokens	observed supported	failed	1	not tested
max tokens	observed supported	degraded	3	invalid input accepted
pairwise core	observed supported	degraded	33	not tested
pairwise core	observed supported	degraded	7	not tested
parameter interaction extended	observed supported	failed	7	not tested
parameter interaction penalties	observed supported	failed	7	not tested
parameter interaction temperature top p output	observed supported	degraded	6	not tested
top logprobs	observed supported	failed	7	not tested

Context and output envelope

Envelope	Documented	Observed functional / realized	Censoring
prompt context window	65.5K	166.7K	right censored
combined prompt + requested output	65.5K	60.1K	right censored
output limit	65.5K	4.1K	—
vision	—	—	—
tool calling	—	—	—
structured output	—	—	—

Deployment guidance

Current gate: /v1/models access recovered after the prepaid balance was replenished; pass two fresh serial streamed controls before deployment or another load wave. Then feature-gate from observed evidence, not the family name. Start below the relevant measured anchor, keep a separate concurrency ceiling, and use open-loop AIMD so rising latency cannot hide offered load. Back off on 429; retry bounded 5xx/timeouts; never retry an unchanged validation error. Re-run the exact profile after a model version, region, quota, or serving-stack change.

Endpoint 06 · MiMo V2.5 Pro

mimo-v2.5-pro

2.1K	115	101	16/16
REQUESTS	LOAD EPOCHS	ERRORS	RESOLVED DIMENSIONS

Inventory

Provider	API	Documented context	Documented output	Input \$/M	Output \$/M
Xiaomi	chat_completions / v1	262.1K	262.1K	\$0.400	\$1.500

Documented modalities: text. Documented tools: True. Structured output: True. Prompt caching: True.

Measured operating anchors

Workload	AIMD healthy offered	Epochs	2-min achieved RPM (95% CI)	2-min status
Short input / short output	≥ 480 RPM	3	24.0 [24.0, 24.0]	measured; composite failed
32K input / short output	—	—	22.9 [20.7, 25.1]	measured; composite failed
Short input / long output	—	—	10.0 [5.6, 14.4]	measured; composite failed
Heterogeneous mixed	—	—	8.7 [7.2, 10.2]	measured; composite failed

Interpretation: the AIMD column is the highest confirmed healthy offered-rate lower bound in the matched workload. The two-minute column is achieved goodput, including drain. Neither is a contractual quota or a diurnal guarantee.

MiMo V2.5 Pro · capability and limits

Capability and validation evidence

Capability / parameter	Transport	Functional	Valid n	Malformed validation
vision	observed transport degraded	not scored	1	not tested
tools	observed supported	passed	2	not tested
parallel tool calls	observed supported	passed	2	not tested
temperature	observed supported	degraded	7	not tested
top p	observed supported	degraded	7	not tested
stop	observed supported	failed	1	not tested
seed	observed supported	passed	2	not tested
caching option	documented unavailable	not scored	0	not tested
capability smoke	observed supported	passed	1	not tested
max completion tokens	observed supported	failed	1	not tested
max tokens	observed supported	degraded	3	correct rejection observed
pairwise core	observed supported	degraded	33	not tested
pairwise core	observed supported	degraded	7	not tested
parameter interaction extended	observed supported	failed	7	not tested
parameter interaction penalties	observed supported	degraded	7	not tested
parameter interaction temperature top p output	observed supported	degraded	6	not tested
response format	observed supported	passed	2	not tested
tool argument bytes	observed supported	degraded	4	not tested

Context and output envelope

Envelope	Documented	Observed functional / realized	Censoring
prompt context window	262.1K	245.5K	nonmonotonic inconclusive
combined prompt + requested output	262.1K	246.3K	right censored
output limit	262.1K	4.1K	—
vision	—	—	—
tool calling	—	—	—
structured output	—	—	—

Deployment guidance

Current gate: /v1/models access recovered after the prepaid balance was replenished; pass two fresh serial streamed controls before deployment or another load wave. Then feature-gate from observed evidence, not the family name. Start below the relevant measured anchor, keep a separate concurrency ceiling, and use open-loop AIMD so rising latency cannot hide offered load. Back off on 429; retry bounded 5xx/timeouts; never retry an unchanged validation error. Re-run the exact profile after a model version, region, quota, or serving-stack change.

Endpoint 07 · Nemotron 3 Ultra

nemotron-3-ultra-550b

1.2K	122	28	16/16
REQUESTS	LOAD EPOCHS	ERRORS	RESOLVED DIMENSIONS

Inventory

Provider	API	Documented context	Documented output	Input \$/M	Output \$/M
NVIDIA	chat_completions / v1	131.1K	131.1K	\$0.900	\$1.700

Documented modalities: text. Documented tools: None. Structured output: None. Prompt caching: None.

Measured operating anchors

Workload	AIMD healthy offered	Epochs	2-min achieved RPM (95% CI)	2-min status
Short input / short output	—	—	59.9 [59.8, 60.1]	measured; composite failed
32K input / short output	≥ 48 RPM	3	44.6 [40.8, 48.3]	measured; composite failed
Short input / long output	—	—	24.5 [21.4, 27.6]	measured; composite failed
Heterogeneous mixed	≥ 12 RPM	3	19.9 [15.5, 24.3]	measured; composite failed

Interpretation: the AIMD column is the highest confirmed healthy offered-rate lower bound in the matched workload. The two-minute column is achieved goodput, including drain. Neither is a contractual quota or a diurnal guarantee.

Nemotron 3 Ultra · capability and limits

Capability and validation evidence

Capability / parameter	Transport	Functional	Valid n	Malformed validation
vision	observed transport degraded	not scored	1	not tested
tools	observed supported	passed	2	not tested
parallel tool calls	observed supported	degraded	2	not tested
temperature	observed supported	passed	7	not tested
top p	observed supported	passed	7	not tested
stop	observed supported	failed	1	not tested
seed	observed supported	passed	2	not tested
caching option	documented unavailable	not scored	0	not tested
capability smoke	observed supported	passed	1	not tested
max completion tokens	observed supported	passed	1	not tested
max tokens	observed supported	degraded	3	correct rejection observed
pairwise core	observed supported	degraded	33	not tested
pairwise core	observed supported	degraded	7	not tested
parameter interaction extended	observed supported	failed	7	not tested
parameter interaction penalties	observed supported	passed	7	not tested
parameter interaction temperature top p output	observed supported	passed	6	not tested
top logprobs	observed supported	passed	7	not tested

Context and output envelope

Envelope	Documented	Observed functional / realized	Censoring
prompt context window	131.1K	76.7K	right censored
combined prompt + requested output	131.1K	122.1K	right censored
output limit	131.1K	4.1K	—
vision	—	—	—
tool calling	—	—	—
structured output	—	—	—

Deployment guidance

Current gate: /v1/models access recovered after the prepaid balance was replenished; pass two fresh serial streamed controls before deployment or another load wave. Then feature-gate from observed evidence, not the family name. Start below the relevant measured anchor, keep a separate concurrency ceiling, and use open-loop AIMD so rising latency cannot hide offered load. Back off on 429; retry bounded 5xx/timeouts; never retry an unchanged validation error. Re-run the exact profile after a model version, region, quota, or serving-stack change.

Endpoint 08 · Nemotron 3 Super

nvidia-nemotron-3-super-120b

1.2K REQUESTS	106 LOAD EPOCHS	442 ERRORS	16/16 RESOLVED DIMENSIONS
-------------------------	---------------------------	----------------------	-------------------------------------

Inventory

Provider	API	Documented context	Documented output	Input \$/M	Output \$/M
NVIDIA	chat_completions / v1	1.00M	32.8K	\$0.300	\$0.650

Documented modalities: text. Documented tools: None. Structured output: None. Prompt caching: True.

Measured operating anchors

Workload	AIMD healthy offered	Epochs	2-min achieved RPM (95% CI)	2-min status
Short input / short output	≥ 60 RPM	3	57.1 [55.5, 58.6]	measured; composite failed
32K input / short output	—	—	22.9 [22.6, 23.2]	measured; composite failed
Short input / long output	≥ 12 RPM	3	1.5 [-0.3, 3.4]	measured; composite failed
Heterogeneous mixed	—	—	—	not complete

Interpretation: the AIMD column is the highest confirmed healthy offered-rate lower bound in the matched workload. The two-minute column is achieved goodput, including drain. Neither is a contractual quota or a diurnal guarantee.

Nemotron 3 Super · capability and limits

Capability and validation evidence

Capability / parameter	Transport	Functional	Valid n	Malformed validation
vision	observed transport degraded	not scored	1	not tested
tools	observed transport degraded	not scored	2	not tested
parallel tool calls	observed transport degraded	not scored	2	not tested
temperature	observed supported	degraded	7	not tested
top p	observed supported	passed	7	not tested
stop	observed supported	failed	1	not tested
seed	observed supported	passed	2	not tested
caching option	documented unavailable	not scored	0	not tested
capability smoke	observed supported	passed	1	not tested
max completion tokens	observed supported	passed	1	not tested
max tokens	observed supported	degraded	3	invalid input accepted
pairwise core	observed supported	degraded	33	not tested
pairwise core	observed supported	passed	7	not tested
parameter interaction extended	observed supported	failed	7	not tested
parameter interaction penalties	observed supported	passed	7	not tested
parameter interaction temperature top p output	observed supported	degraded	6	not tested
top logprobs	observed supported	passed	7	not tested

Context and output envelope

Envelope	Documented	Observed functional / realized	Censoring
prompt context window	1.00M	76.7K	interval censored
combined prompt + requested output	1.00M	—	left censored
output limit	32.8K	4.1K	—
vision	—	—	—
tool calling	—	—	—
structured output	—	—	—

Deployment guidance

Current gate: /v1/models access recovered after the prepaid balance was replenished; pass two fresh serial streamed controls before deployment or another load wave. Then feature-gate from observed evidence, not the family name. Start below the relevant measured anchor, keep a separate concurrency ceiling, and use open-loop AIMD so rising latency cannot hide offered load. Back off on 429; retry bounded 5xx/timeouts; never retry an unchanged validation error. Re-run the exact profile after a model version, region, quota, or serving-stack change.

Endpoint 09 · GPT-OSS 120B

openai-gpt-oss-120b

9.1K	144	37	16/16
REQUESTS	LOAD EPOCHS	ERRORS	RESOLVED DIMENSIONS

Inventory

Provider	API	Documented context	Documented output	Input \$/M	Output \$/M
OpenAI	chat_completions / v1	128.0K	4.1K	\$0.100	\$0.700

Documented modalities: text. Documented tools: None. Structured output: None. Prompt caching: True.

Measured operating anchors

Workload	AIMD healthy offered	Epochs	2-min achieved RPM (95% CI)	2-min status
Short input / short output	≥ 1.9K RPM	3	692 [12.7, 1.4K]	measured; composite failed
32K input / short output	≥ 144 RPM	3	129 [121, 137]	measured; composite failed
Short input / long output	—	—	10.6 [7.4, 13.7]	measured; composite failed
Heterogeneous mixed	≥ 96 RPM	3	15.9 [10.3, 21.6]	measured; composite failed

Interpretation: the AIMD column is the highest confirmed healthy offered-rate lower bound in the matched workload. The two-minute column is achieved goodput, including drain. Neither is a contractual quota or a diurnal guarantee.

GPT-OSS 120B · capability and limits

Capability and validation evidence

Capability / parameter	Transport	Functional	Valid n	Malformed validation
vision	observed transport degraded	not scored	1	not tested
tools	observed supported	passed	2	not tested
parallel tool calls	observed supported	degraded	2	not tested
temperature	observed supported	passed	7	not tested
top p	observed supported	degraded	7	not tested
stop	observed supported	failed	1	not tested
seed	observed supported	failed	2	not tested
caching option	documented unavailable	not scored	0	not tested
capability smoke	observed supported	passed	1	not tested
max completion tokens	observed supported	passed	1	not tested
max tokens	observed supported	degraded	3	invalid input accepted
pairwise core	observed supported	degraded	33	not tested
pairwise core	observed supported	degraded	7	not tested
parameter interaction extended	observed supported	failed	7	not tested
parameter interaction penalties	observed supported	degraded	7	not tested
parameter interaction temperature top p output	observed supported	passed	6	not tested
top logprobs	observed supported	degraded	7	not tested

Context and output envelope

Envelope	Documented	Observed functional / realized	Censoring
prompt context window	128.0K	66.7K	right censored
combined prompt + requested output	128.0K	113.9K	right censored
output limit	4.1K	32.8K	—
vision	—	—	—
tool calling	—	—	—
structured output	—	—	—

Deployment guidance

Current gate: /v1/models access recovered after the prepaid balance was replenished; pass two fresh serial streamed controls before deployment or another load wave. Then feature-gate from observed evidence, not the family name. Start below the relevant measured anchor, keep a separate concurrency ceiling, and use open-loop AIMD so rising latency cannot hide offered load. Back off on 429; retry bounded 5xx/timeouts; never retry an unchanged validation error. Re-run the exact profile after a model version, region, quota, or serving-stack change.

Endpoint 10 · Qwen 3.8 Max

qwen3.8-max

1.5K	117	55	16/16
REQUESTS	LOAD EPOCHS	ERRORS	RESOLVED DIMENSIONS

Inventory

Provider	API	Documented context	Documented output	Input \$/M	Output \$/M
Alibaba	chat_completions / v1	1.00M	262.1K	\$2.000	\$6.000

Documented modalities: text, image, video. Documented tools: True. Structured output: True. Prompt caching: True.

Measured operating anchors

Workload	AIMD healthy offered	Epochs	2-min achieved RPM (95% CI)	2-min status
Short input / short output	—	—	59.1 [56.2, 62.0]	composite passed
32K input / short output	—	—	11.8 [11.1, 12.5]	measured; composite failed
Short input / long output	—	—	23.1 [20.5, 25.8]	measured; composite failed
Heterogeneous mixed	≥ 12 RPM	3	6.0 [6.0, 6.0]	measured; composite failed

Interpretation: the AIMD column is the highest confirmed healthy offered-rate lower bound in the matched workload. The two-minute column is achieved goodput, including drain. Neither is a contractual quota or a diurnal guarantee.

Qwen 3.8 Max · capability and limits

Capability and validation evidence

Capability / parameter	Transport	Functional	Valid n	Malformed validation
vision	observed supported	passed	18	correct rejection observed
tools	observed supported	passed	2	not tested
parallel tool calls	observed supported	passed	2	not tested
temperature	observed supported	passed	7	not tested
top p	observed supported	passed	7	not tested
stop	observed supported	passed	1	not tested
seed	observed supported	passed	2	not tested
caching option	documented unavailable	not scored	0	not tested
capability smoke	observed supported	passed	1	not tested
max completion tokens	observed supported	passed	1	not tested
max tokens	observed supported	degraded	3	invalid input accepted
pairwise core	observed supported	degraded	33	not tested
pairwise core	observed supported	degraded	7	not tested
parameter interaction extended	observed supported	failed	7	not tested
parameter interaction penalties	observed supported	passed	7	not tested
parameter interaction temperature top p output	observed supported	passed	6	not tested
response format	observed supported	passed	2	not tested
tool argument bytes	observed supported	degraded	4	not tested

Context and output envelope

Envelope	Documented	Observed functional / realized	Censoring
prompt context window	1.00M	1.00M	right censored
combined prompt + requested output	1.00M	1.00M	right censored
output limit	262.1K	10.4K	—
vision	—	—	—
tool calling	—	—	—
structured output	—	—	—

Deployment guidance

Current gate: /v1/models access recovered after the prepaid balance was replenished; pass two fresh serial streamed controls before deployment or another load wave. Then feature-gate from observed evidence, not the family name. Start below the relevant measured anchor, keep a separate concurrency ceiling, and use open-loop AIMD so rising latency cannot hide offered load. Back off on 429; retry bounded 5xx/timeouts; never retry an unchanged validation error. Re-run the exact profile after a model version, region, quota, or serving-stack change.

Endpoint 11 · Qwen 3.5 397B

qwen3.5-397b-a17b

4.0K REQUESTS	105 LOAD EPOCHS	1.6K ERRORS	16/16 RESOLVED DIMENSIONS
-------------------------------------	---------------------------------------	-----------------------------------	---

Inventory

Provider	API	Documented context	Documented output	Input \$/M	Output \$/M
Alibaba	chat_completions / v1	131.1K	131.1K	\$0.550	\$3.500

Documented modalities: text. Documented tools: None. Structured output: None. Prompt caching: True.

Measured operating anchors

Workload	AIMD healthy offered	Epochs	2-min achieved RPM (95% CI)	2-min status
Short input / short output	≥ 960 RPM	3	397 [318, 475]	measured; composite failed
32K input / short output	—	—	19.8 [18.4, 21.2]	measured; composite failed
Short input / long output	—	—	2.6 [2.6, 2.7]	measured; composite failed
Heterogeneous mixed	—	—	—	not complete

Interpretation: the AIMD column is the highest confirmed healthy offered-rate lower bound in the matched workload. The two-minute column is achieved goodput, including drain. Neither is a contractual quota or a diurnal guarantee.

Qwen 3.5 397B · capability and limits

Capability and validation evidence

Capability / parameter	Transport	Functional	Valid n	Malformed validation
vision	observed supported	failed	1	not tested
tools	observed transport degraded	not scored	2	not tested
parallel tool calls	observed transport degraded	not scored	2	not tested
temperature	observed supported	failed	7	not tested
top p	observed supported	failed	7	not tested
stop	observed supported	failed	1	not tested
seed	observed supported	failed	2	not tested
caching option	documented unavailable	not scored	0	not tested
capability smoke	observed supported	failed	1	not tested
max completion tokens	observed supported	failed	1	not tested
max tokens	observed supported	failed	3	correct rejection observed
pairwise core	observed supported	degraded	33	not tested
pairwise core	observed supported	failed	7	not tested
parameter interaction extended	observed supported	failed	7	not tested
parameter interaction penalties	observed supported	failed	7	not tested
parameter interaction temperature top p output	observed supported	degraded	6	not tested
top logprobs	observed supported	failed	7	not tested

Context and output envelope

Envelope	Documented	Observed functional / realized	Censoring
prompt context window	131.1K	98.1K	nonmonotonic inconclusive
combined prompt + requested output	131.1K	113.8K	interval censored
output limit	131.1K	5.6K	—
vision	—	—	—
tool calling	—	—	—
structured output	—	—	—

Deployment guidance

Current gate: /v1/models access recovered after the prepaid balance was replenished; pass two fresh serial streamed controls before deployment or another load wave. Then feature-gate from observed evidence, not the family name. Start below the relevant measured anchor, keep a separate concurrency ceiling, and use open-loop AIMD so rising latency cannot hide offered load. Back off on 429; retry bounded 5xx/timeouts; never retry an unchanged validation error. Re-run the exact profile after a model version, region, quota, or serving-stack change.

Incident appendix · excluded partner model

Arcee Trinity is not in the hosted-only production scope

DigitalOcean's current documentation lists Arcee Trinity in a separate Arcee partner-model section, before the DigitalOcean-Hosted Models table. Startup-program credits exclude third-party inference hosted outside DigitalOcean infrastructure. The benchmark should therefore not have included this endpoint under a credits-only instruction.

2.8K HISTORICAL ROWS	\$7.284 TOKEN-ATTRIBUTED ESTIMATE	EXCLUDED PRODUCTION COMPARISONS
--------------------------------	---	---

The rows remain in the immutable evidence bundle for cost reconciliation and forensic reproducibility. They are excluded from every chart, portfolio KPI, endpoint recommendation, and future spend-bearing default in this encyclopedia. The code now uses a documented hosted-model allowlist and rejects an explicit Arcee selection before any request can be sent.

The request ledger alone cannot prove which exact balance line item caused the prepaid account to reach zero. However, this partner-model usage is the campaign's identified passthrough exposure and is the most plausible reason credits did not absorb all inference charges.

9. Measurement contract

Term	Operational definition
Offered RPM	Arrivals scheduled by the open-loop generator per minute. Slow responses do not reduce this denominator.
Achieved RPM	Completed requests divided by full elapsed wall time.
Effective input TPM	Prompt tokens on successful requests divided by full elapsed wall-clock minutes.
Effective output TPM	Completion tokens on successful requests divided by full elapsed wall-clock minutes.
TTFT	Time from send to first streamed content event. Buffered responses are censored.
Post-TTFT output proxy	Billed completion tokens divided by request time minus streamed TTFT. It is not server-internal decoder speed.
AIMD lower bound	Highest offered rate with the required separated healthy confirmation epochs; right-censored if no overload knee was found.
Two-minute interval	Student-t interval over four contiguous 30-second blocks. Exploratory because serial correlation is not modeled.
Quality-adjusted goodput	Goodput multiplied only by scores from deterministic task checks; unscored outputs are not presumed correct.

Outlier policy

No timing value is trimmed, winsorized, or silently clipped. Impossible or unstable denominators are excluded by a deterministic validity rule and retained in metric-audit.csv. Qualified extremes remain in the raw derived table and are flagged. Endpoint charts compare matched workload cells only.

Statistical unit

Serial baselines use independent request IDs. AIMD capacity uses epochs. Two-minute stability uses the four predeclared blocks. Paired quality uses pair IDs. Output tokens are never treated as independent observations. Sparse p99 values are suppressed rather than decorated with false precision.

10. Cost, reproducibility, and reuse

\$272.220 REQUEST-ATTRIBUTED ESTIMATE	\$329.972 AUTHORITATIVE EXPOSURE	\$400.000 CAMPAIGN CAP
---	--	--

The two cost figures overlap and must not be added. The request-attributed estimate shown here covers only the 11 hosted endpoints; historical partner-model rows are reported separately in the incident appendix. Conservative exposure remains the authoritative cumulative guard and retains every earlier stage plus worst-case reservations for failed, timed-out, or usage-incomplete requests.

Reproduce a provider run

Step	Required artifact
1	Freeze exact endpoint IDs, API surface, region, pricing, limits, and capability documentation.
2	Generate the provider-neutral workload matrix and immutable request identities.
3	Run low-load baselines, open-loop AIMD, two-minute candidate soaks, capability probes, and context boundaries in isolated lanes.
4	Persist request, epoch, block, reservation, and cost receipts without prompts, outputs, credentials, or raw headers in the public bundle.
5	Normalize with the shared metric schema; qualify timing; compute intervals at the correct sampling unit; label censoring.
6	Render matched charts and endpoint profiles; run schema, matrix, sample, unit, CI, secret, and visual gates before publication.

What this report does not prove

It does not prove current account availability, 24-hour or regional stability, contractual quotas, server-internal prefill or decoder speed, or causal differences versus another serving provider. It provides auditable tested anchors, failure distributions, feature evidence, context behavior, and reusable experiment code from the completed measurement periods.